



Drinking salt water

A new membrane made out of wood has been developed to filter salt from seawater.
<https://digit.in/sep19-17>



Ghost particles

Scientists have now measured the mass of Neutrinos also known as ghost particles.
<https://digit.in/sep19-18>

**Clouds
plunging into
the supermassive
black hole at
Abell 2597.**

Credit: ESO

GALACTIC WEATHER

Supermassive black holes at the centers of galaxies are responsible for all kinds of stellar weather, including galactic wind, rain, deluges and fountains

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here are streams of high energy particles emanating from the Sun, depending on how active it is. This

is known as solar wind. A similar phenomenon occurs on a much larger scale. High energy particles stream out of galaxies at speeds ranging between 300 to 3,000 kilometers an hour. The material is ejected into a roughly spherical region around galaxies, known as the halo. The halo around the Milky Way was detected by the ROSAT X-ray telescope in the early 1990s, and such halos have been found around other spiral galaxies as well. At times, the material can even escape the

galaxy entirely, and escape into the intergalactic medium.

These winds are generated by newly-formed stars, or stars dying in supernova events, both of which result in highly energetic and dense outpourings of particles. Active galactic nuclei, or supermassive black holes at the centers of galaxies, can also be a source of galactic wind. Galaxies with a large number of newly-forming stars are called starburst galaxies, and have visible clouds or jets of gas on two sides. One of the most well-known examples of such a galaxy is M82, with a visible cloud of winds extending to around 1000 light years on either side. The outpouring of material from another galaxy, NGC

4666 is so violent, that it is termed as superwind. The material jets extend for tens of thousands of light years, and are believed to be originating from a combination of vigorous star formation, supernovae and massive stars. The gas particles that make up the wind emit strong X-ray radiation, which is not visible to the naked eye. This radiation cools down the material, causing them to fall back towards the plane of the galaxy. Galactic wind circulate the raw material of the stars and planets, help redistribute metals across the planets, and also play a role in the regulation of star formation.

About a billion light years away from the Earth, in the constellation of Aquarius is one of the largest known structures in the universe, designated as Abell 2597. This is a cluster of over fifty galaxies, surrounding a giant elliptical galaxy with a supermassive black hole at its core. Scientists observing Abell 2597, using two astronomical instruments, the Atacama Large Millimeter /submillimeter Array (ALMA) and the Multi-Unit Spectroscopic Explorer (MUSE) on board the Very Large Telescope (VLT), found something incredibly surprising. The gases were being ejected by the galactic wind, but unable to escape the gravity of the galaxy, they were falling back in towards the supermassive black hole at the center. The influx of gas into the accretion disk around a